

TOWARDS RELIABLE EVENT PROCESSING
IN SYSTEMS OF TWINNED SYSTEMS

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TWINNED SYSTEMS

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Abstract

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Notation, Definitions, and Abbreviations

CPS	Cyber-Physical System
CEP	Complex Event Processing
DT	Digital Twin
SoS	System of Systems
SoTS	System of Twinned Systems
FDI	Fault Detection and Isolation
LTL	Linear Temporal Logic
CTL	Computation Tree Logic
NSGA-II	Non-dominated Sorting Genetic Algorithm II
DBN	Dynamic Bayesian Network
LSTM	Long Short-Term Memory

FDNA	Functional Dependency Network Analysis
FMEA	Failure Mode and Effects Analysis
FMECA	Failure Mode, Effects, and Criticality Analysis
FTA	Fault Tree Analysis
ETA	Event Tree Analysis
HAZOP	Hazard and Operability Study
CTMM	Continuous-Time Markov Model
SHSA	Substitutable Heterogeneous System Architecture

Declaration of Academic Achievement

TODO, add last - The student will declare his/her research contribution and, as appropriate, those of colleagues or other contributors to the contents of the thesis.

Part I

Reliability in Systems of Twinned Systems

Chapter 1

Reliability Through Orthogonal Lifecycle Modeling in Systems of Twinned Systems

This chapter reframes reliability in SoTS at the mission level. It introduces belonging and health as primary disturbance dimensions and formalizes their interaction through an orthogonal lifecycle modeling framework. Within this framework, escalating levels of reliability impose progressively stricter architectural constraints on participation to ensure reliable SoTS operation.

Example Distributed CEP in a SoTS:

Throughout this chapter, we consider a distributed SoTS composed of multiple autonomous DTS, each associated with an operational constituent.

Each constituent continuously produces time-stamped events representing measurements, state transitions, or operational observations. These event streams are transmitted to a higher-level CEP layer responsible for aggregating, correlating, and interpreting distributed information.

The CEP layer detects higher-order patterns by applying temporal and logical constraints over combinations of constituent-level events. These inferred composite events represent mission-relevant situations that are not observable at the level of any single constituent.

The correctness and timeliness of mission-level inference therefore depend on the collective availability and quality of the incoming event streams. This event-aggregation architecture provides a concrete setting for the formalization developed in the remainder of the thesis.

1.1 How Disturbances Impact Reliability

In dependability theory, reliability is defined as the ability to deliver correct and continuous service [1]. However, SoS differ from traditional systems due to the operational and managerial independence of their constituents [2]. Because constituents are autonomous and may join, leave, or behave independently, reliability cannot be reduced to component-level correctness. Instead, reliability must be understood at the mission level.

In this context, a disturbance is any structural or behavioral change arising from

constituent interdependence, evolution, or undesired emergent behavior that affects capability availability or mission performance. Following [3], reliability in SoS is therefore characterized as the ability of the system to tolerate such disturbances while continuing to fulfill its mission.

In SoTS, this challenge is amplified. Each operational constituent is paired with a digital counterpart, so disturbances propagate across both physical and digital layers. Structural reconfiguration, performance degradation, or behavioral deviation compromise synchronization, representation validity, and data-driven decision-making.

Two primary disturbance dimensions emerge.

Dimension 1 — Belonging. Participation in an SoS is voluntary and autonomous. Constituents may join or leave at will, introducing structural disturbances that affect capability availability and mission execution [4]. A constituent may therefore be:

- *Belongs*, structurally present and contributing its capabilities to the SoTS; or
- *Does not Belong*, structurally absent and unable to contribute to mission execution.

Changes in belonging alter the set of available capabilities and may disrupt dependent activities.

Dimension 2 - Health. Even when structurally present, a constituent may degrade, misalign, or underperform relative to mission expectations [5]. In dependability theory, such degradation arises from faults that activate into errors and may lead to failure. A constituent may therefore be:

- *Healthy*, capable of delivering correct and continuous service in alignment with its expected role; or
- *Unhealthy*, affected by faults, errors, or performance degradation that undermine reliable operation.

Disturbances thus arise not only from changes in membership, but also from degradation within participating constituents.

1.1.1 The Belonging-Health Disturbance Space

Belonging and health together define a two-dimensional disturbance space.

	Healthy	Unhealthy
Belongs	Normal participation. Structurally present and operationally reliable.	Degraded participation. Structurally present but operationally unreliable, potentially propagating disturbances.
Does not belong	Capability loss. Operationally capable but absent, disrupting dependent constituents.	Externalized disturbance. Absent and degraded; instability lies outside the SoTS

Each quadrant represents a distinct disturbance configuration. Because disturbances arise from both belonging and health, reliability in SoTS must account for both dimensions. Ensuring reliability therefore requires architectural regulation of participation across them. The following section introduces a formal mechanism to achieve this regulation.

1.2 Orthogonal Lifecycle Model for Participation Regulation

Although constituent belonging and participation have been extensively studied in SoS research, and reliability has been formally defined in dependability theory [6], no unified framework links operational condition to participation in SoTS.

To address this gap, I propose an orthogonal lifecycle modeling framework that couples belonging and health as parallel state dimensions. Lifecycle transitions are conditioned on operational state, modeling participation and health as interacting regions. Within this framework, escalating reliability levels impose progressively stricter architectural constraints on participation: lower levels tolerate disturbances reactively, whereas higher levels proactively restrict, delay, or prevent participation when health conditions threaten SoTS reliability.

The formalization below is grounded in the motivating SoTS use case ?? presented earlier. While the specific states reflect that scenario, the orthogonal modeling structure is domain-agnostic and can be instantiated with alternative belonging, health, and admissibility definitions as required.

1.2.1 Orthogonal State Composition

Participation is modeled over two orthogonal dimensions: belonging and health.

Belonging Dimension. Grounded in the constituent lifecycle model proposed by [7], the initial set of belonging states is defined as:

$$B = \{Disengaged, Prepared, Passive, Active\}$$

- *Disengaged* (renamed from *Ignorant*): the constituent does not structurally participate in the SoTS.
- *Prepared*: the constituent is capable of participation but not yet structurally engaged.
- *Passive*: the constituent participates with limited coordination or influence.
- *Active*: the constituent is fully integrated and actively coordinated within the SoTS.

Health Dimension. Grounded in [5] multi-level reliability model, system impairment is characterized through seven operational states representing progressive degradation:

$$H = \{Ideal, Defective, Faulty, Erroneous, Malfunctioning, Degraded, Failed\}$$

- *Ideal*: no defects or impairments are present.
- *Defective*: latent defects exist but have not yet activated.
- *Faulty*: active faults are present internally.
- *Erroneous*: internal state corruption has occurred.
- *Malfunctioning*: incorrect behavior is externally observable.
- *Degraded*: service is delivered with reduced capability.
- *Failed*: required service can no longer be delivered.

Orthogonal Composition. The complete participation state of a constituent is defined as the Cartesian product:

$$S = B \times H$$

Each composite state $(b, h) \in S$ represents a distinct disturbance configuration. Escalating reliability levels introduce additional states and progressively restrict the admissible products and transitions between these dimensions.

1.2.2 Reliability Levels

I now present the state charts corresponding to each reliability level. Each level extends the previous one by progressively strengthening architectural control over participation and constraining the admissible combinations of lifecycle belonging and operational health.

Level 1 - Reactive

Level 1 reflects the most loosely coupled and least regulated form of SoTS. Following [7], constituents are required to make participation changes visible to the SoS. We instantiate this architecturally by enforcing mandatory *announcement* actions on all lifecycle transitions, including joining or leaving the SoS and joining or leaving a constellation, as shown in Figure 1.1. At this level, the two regions evolve independently. No architectural guards regulate admissible combinations of health and belonging.

This represents the weakest form of architectural control. Participation is made observable through mandatory announcements, but operational health does not regulate admission or continued involvement.

Such a configuration aligns naturally with highly collaborative or virtual SoTS, where constituents retain autonomy and central governance is minimal. The SoTS therefore operates reactively, responding to disturbances only after they become externally visible.

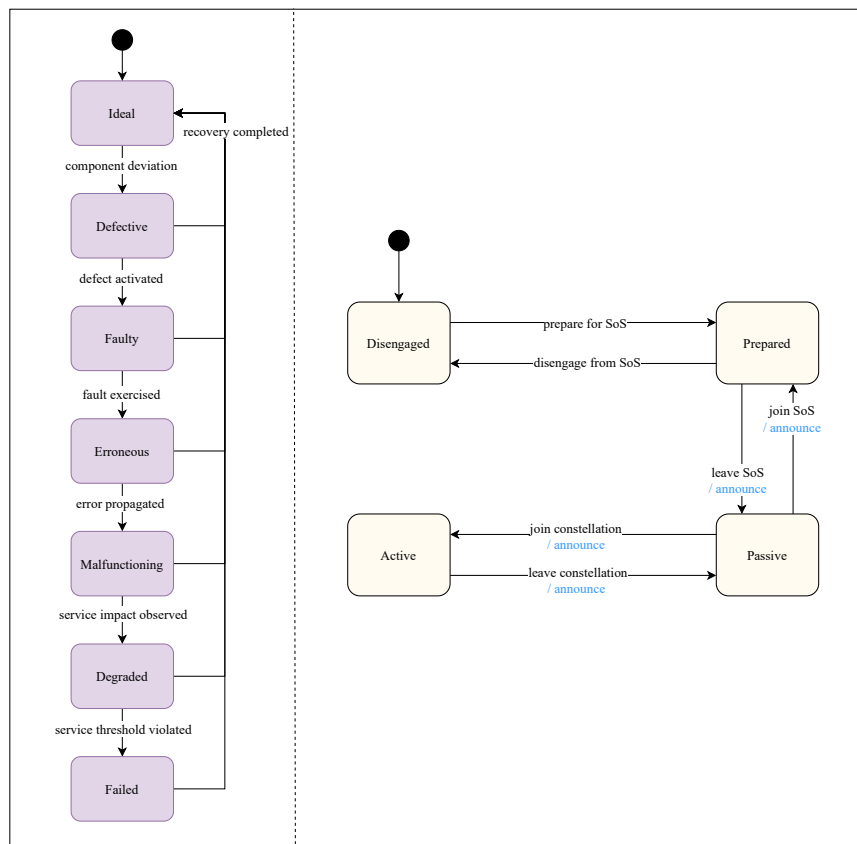


Figure 1.1: Level 1: Reactive

Level 2 - Proactive

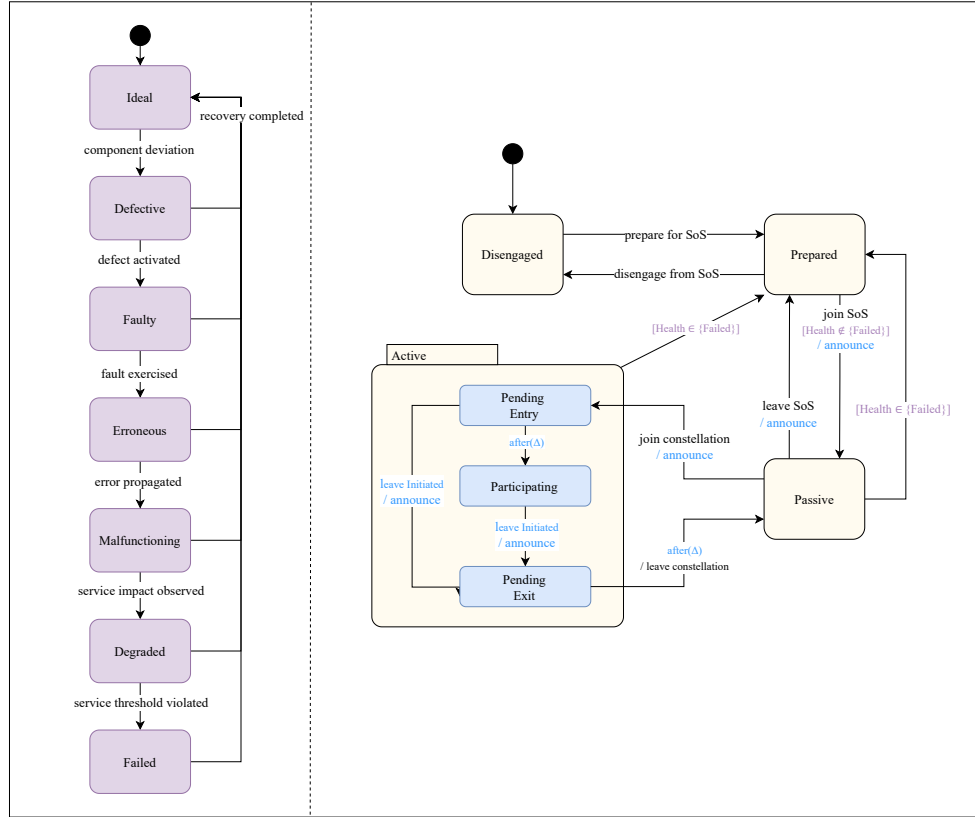


Figure 1.2: Level 2: Proactive

Level 2 introduces reliability-aware governance by coupling operational health to participation. Mandatory *announcement* actions remain enforced on all lifecycle transitions, but health now constrains admissible combinations of belonging and activity, as shown in Figure 1.2

If $Health \in Failed$ while in the *Active* or *Passive* state, the architecture initiates an exit sequence. This ensures that failed constituents cannot remain actively participating in a constellation. Admission is also guarded: the transition *join SoS* is permitted only when $Health \notin Failed$, preventing failed constituents from entering

the system.

The *Active* state is refined into three substates: *Pending Entry*, *Participating*, and *Pending Exit*. Temporal delays (e.g., *after*(Δ)) are present during transitions from *Pending Entry* $\rightarrow$ *Participating* and from *Pending Exit* $\rightarrow$ *Passive*. These delays assist with disturbance handling. During entry, the delay allows a constituent to stabilize before fully participating. During exit, the delay ensures withdrawal is controlled rather than abrupt, reducing the risk of sudden disruption to other participants.

This level aligns most naturally with *Directed* and *Acknowledged SoTS*, where centralized oversight exists and basic containment of disturbances is required, and is less typical of highly autonomous *Collaborative* or *Virtual SoTS*.

Level 3 - Controlled

Level 3 strengthens architectural regulation by introducing explicit negotiation and tighter participation control. Mandatory *announcement* actions remain enforced, and this level includes all health-based constraints and stabilization mechanisms introduced in Level 2.

Admission to a constellation is now mediated through a negotiation phase within the *Passive* state, which is refined into *Available* and *Negotiating*. A constituent may transition into negotiation depending on its health condition. If issuing a *join request*, the constituent must be in ideal health ($Health \in Ideal$). If receiving a *join invitation*, a constituent may be admitted in an operational but not perfect state ($Health \in Ideal, Defective, Faulty$), reflecting scenarios where the SoTS may require additional capacity despite minor impairments.

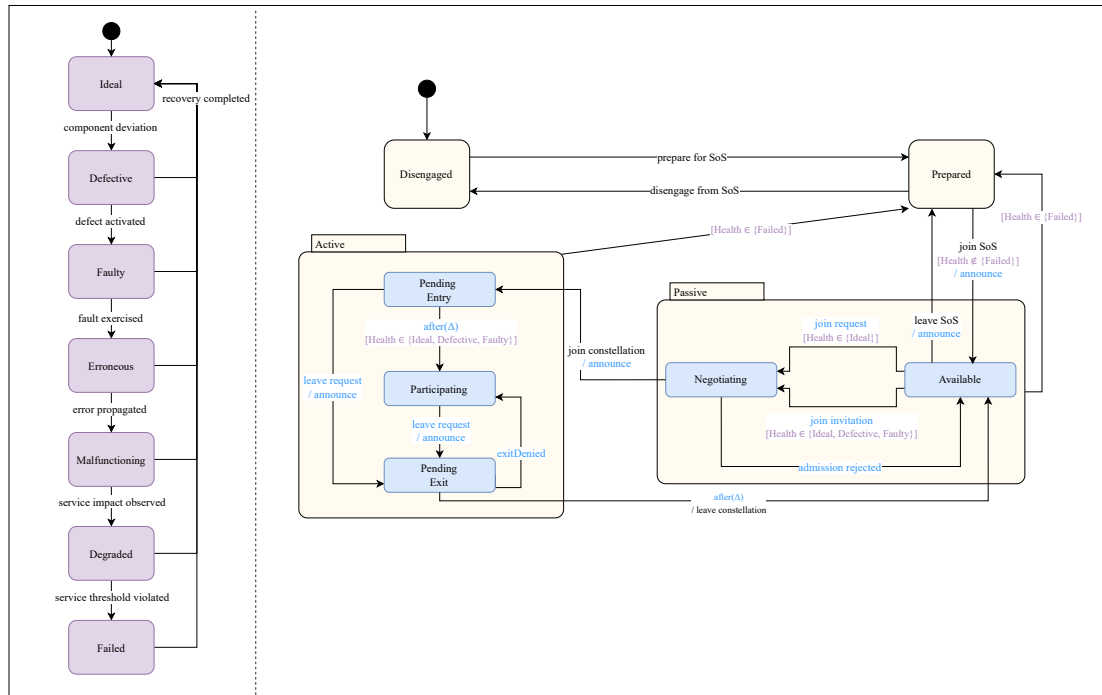


Figure 1.3: Level 3: Controlled

Active participation remains structured through *Pending Entry*, *Participating*, and *Pending Exit*. However, exit is no longer voluntary. A constituent must issue a *leave request*, and departure may be denied (*exitDenied*) if architectural conditions are not satisfied. This prevents destabilizing during critical phases.

Unlike Level 2, participation is no longer only health-constrained. It is also regulated through negotiation and controlled admission and exit decisions. Lifecycle evolution becomes an explicitly governed process. This level aligns most naturally with *Directed SoTS* and strongly governed *Acknowledged SoTS*, where coordinated control and regulated participation are required.

Level 4 - Adaptive

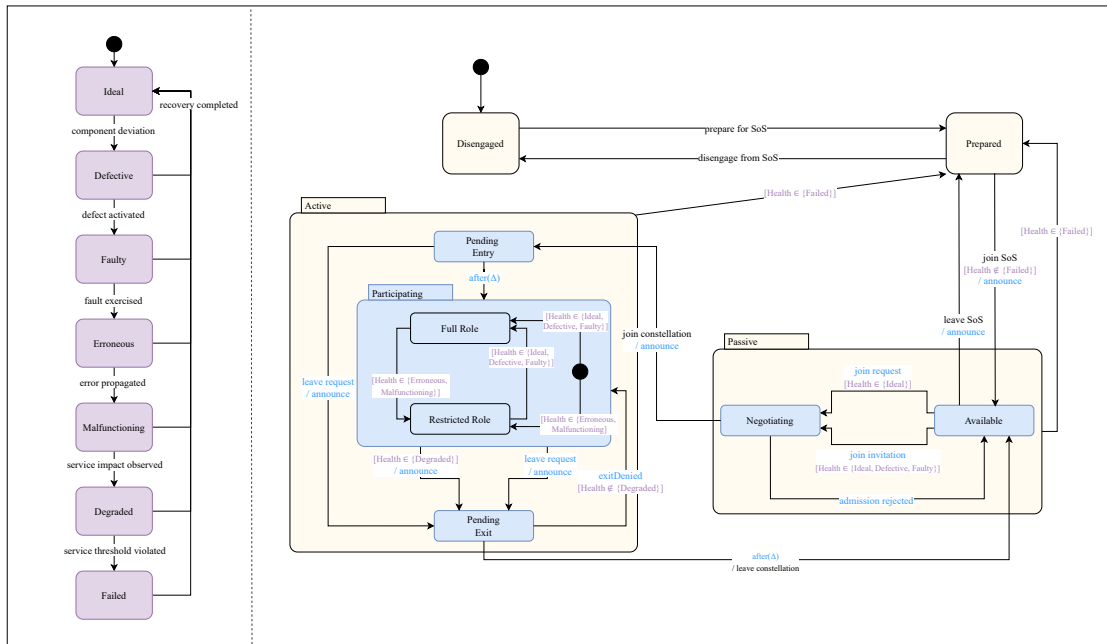


Figure 1.4: Level 4: Adaptive

Level 4 extends Level 3 by introducing role adaptation based on operational health.

All enforcement mechanisms from the previous levels remain in place: mandatory announcements, guarded admission, negotiated entry, controlled exit, stabilization delays, and failure-triggered disengagement.

The key refinement occurs within the *Participating* state. The state is decomposed into two roles: *Full Role* and *Restricted Role*. Health now determines not only whether a constituent may participate, but how it participates.

When $Health \in \text{Ideal, Defective, Faulty}$, the constituent may operate in a *Full Role*. If health degrades to $Health \in \text{Erroneous, Malfunctioning}$, participation is downgraded to a *Restricted Role*. In this mode, the constituent remains part of the constellation but operates with reduced responsibility, limiting the potential propagation of disturbance.

Level 4 no longer waits for complete failure to enforce exit. When $Health \in \text{Degraded}$, the architecture proactively initiates a leave request, triggering a controlled withdrawal with proper notice. Failure still results in enforced disengagement, but intervention now occurs earlier to limit disturbance propagation.

Level 4 enables adaptive containment: degraded constituents can be functionally reconfigured before removal. Disturbance handling therefore shifts from reactive containment (Level 2) and negotiated control (Level 3) to dynamic role reconfiguration.

This level aligns most naturally with strongly governed *Directed SoTS*, and advanced *Acknowledged SoTS*, where architectural intelligence, often embodied in a coordinating Digital Twin, can assess health and reassign responsibilities in real time.

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